Project Check-in 2

Sketch-to-Photo Synthesis with ControlNet-Guided Diffusion

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Team deliverable

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1. Repository and invitations

All project code and this sequence of check-ins are in one GitHub repository:

https://github.com/LRozay/Sketch-to-Photo

The instructor (bertin.cordovadiba@sjsu.edu) and the TA (charansai.gandham@sjsu.edu) have been invited to this repository. Section 8 lists the nine meaningful check-ins already on *main*.

This document is the initial draft of the final report. It contains the tentative abstract, the report outline, the literature survey, the approach we are developing, the algorithms in that pipeline, and the repository record required for Check-in 2.

2. Finalized specifications

The scope is unchanged from the proposal and from Check-in 1. We generate a photorealistic image from a sketch and a short text prompt. The baseline is Stable Diffusion 1.5 with the pretrained ControlNet canny and scribble checkpoints, loaded through Hugging Face Diffusers. We then adapt ControlNet with LoRA on a small set of real hand-drawn sketches while Stable Diffusion 1.5 stays frozen. The same sketches are used to compare the pretrained baseline with the adapted model. We ablate the ControlNet conditioning scale and the classifier-free guidance scale. Photos and conditioning images are resized to 512×512. Training and batch inference are planned for the College of Engineering HPC machines, because latent diffusion at this resolution needs a GPU.

Measurement will use three views of quality. FID compares generated images with real COCO photographs. CLIP score measures agreement with the text prompt. A small pairwise human survey asks which image better follows the sketch. The repository currently holds the pipeline code. It does not yet hold FID numbers, CLIP numbers, or survey responses. Those measurements are the next experimental step, and we do not report invented scores here.

3. Tentative abstract

Rough sketches are a natural way to specify a picture during early design, but pretrained ControlNet models are strongest on clean edge maps. Real freehand drawings have broken strokes, uneven pressure, and missing detail. This project asks how far a Stable Diffusion 1.5 plus ControlNet pipeline can move from synthetic edges to that kind of drawing when only a small set of real sketches is available for adaptation.

We use a COCO 2017 subset of about 2,000 to 5,000 captioned photographs as the source of photo targets

and text prompts. Synthetic conditioning images are Canny and HED edge maps of those photographs, following the paired-edge setup used to train ControlNet. A separate set of 50 to 100 hand-drawn sketches, each with a short prompt and a target photo, is the fine-tuning and test set. Stable Diffusion 1.5 remains frozen. LoRA adapters are trained on the ControlNet attention projections so the model can follow messier strokes without a full retraining of the diffusion network.

The comparison is between the pretrained canny and scribble baselines and the LoRA-adapted ControlNet on the same sketches and prompts. Conditioning scale and guidance scale are swept because both change the balance between sketch fidelity and photorealism. We will report FID, CLIP score, and a human preference survey, together with side-by-side failures. The inference path, the LoRA training path, the scale sweep, and the hand-drawn data check are implemented. Running them on the collected sketches and recording the scores are the remaining experimental steps.

4. Outline of the final report

The final report is planned in the following order. This check-in drafts the abstract, the related-work survey, the approach, and the algorithms, and it states the data and evaluation design.

1. **Abstract.** Problem, pipeline, comparison, and metrics.
2. **Introduction and motivation.** Why sketch-to-photo matters for early design, and why clean edge maps are a weak stand-in for freehand drawings.
3. **Related work.** Denoising diffusion, latent diffusion and Stable Diffusion, ControlNet, LoRA, edge detection for synthetic sketches, and the metrics we will use. This check-in contains that survey.
4. **Data.** COCO 2017 subset, captions used as prompts, Canny and HED pairing, and the 50 to 100 hand-drawn sketches.
5. **Method.** Frozen Stable Diffusion 1.5, pretrained canny and scribble ControlNets, and LoRA on the ControlNet attention layers.
6. **Experiments and evaluation setup.** Baseline versus fine-tune, conditioning-scale and guidance-scale grid, FID, CLIP score, and the pairwise survey.
7. **Results.** Side-by-side grids on identical sketches. This section will be filled after the HPC runs. It is intentionally empty of numbers in this draft.
8. **Discussion and failure cases.** Where the baseline ignores thin strokes, where LoRA overfits a small sketch set, and where the prompt fights the sketch.
9. **Conclusion and future work.** What transferred from synthetic edges to freehand input, and what a larger sketch set or a different adapter rank would change.
10. **References.**

5. Literature survey

The pipeline sits on four published ideas: denoising diffusion, latent diffusion, spatial control of a frozen diffusion model, and low-rank adaptation. Edge detectors and the two evaluation metrics are part of the same design, so they are included here.

5.1 Denoising diffusion

Ho, Jain, and Abbeel (2020) define a diffusion model as a fixed forward process that adds Gaussian noise

to an image over many steps, and a learned reverse process that removes that noise. The usual training loss predicts the noise that was added at a randomly chosen timestep. Sampling starts from pure noise and applies the learned denoiser repeatedly. That objective is the training loss in our LoRA script: the frozen U-Net predicts noise in the latent of the target photo, and the loss is mean squared error against the noise that was added. The pretrained sampler we use at inference is UniPC with 30 steps, which is a fast solver for the same kind of reverse process. The training noise schedule itself stays the Stable Diffusion DDPM scheduler.

5.2 Latent diffusion and Stable Diffusion 1.5

Rombach et al. (2022) run the diffusion process in the latent space of a pretrained autoencoder instead of in pixel space. The autoencoder compresses a 512 × 512 image by a factor of eight, so the U-Net denoises a 64 × 64 latent. A CLIP text encoder turns the prompt into a sequence of embeddings that the U-Net attends to, and classifier-free guidance scales how strongly that text pulls the sample. Stable Diffusion 1.5 (*runwayml/stable-diffusion-v1-5*) is the checkpoint we freeze. In training we encode the target photo with its variational autoencoder, scale the latent by the autoencoder's scaling factor, add noise, and leave the text encoder, the autoencoder, and the U-Net without gradients. This is why a small sketch set is usable at all: the photorealistic prior is already in the frozen model, and we do not have to learn appearance from 50 to 100 drawings.

5.3 ControlNet

Zhang, Rao, and Agrawala (2023) add a spatial condition, such as an edge map, a scribble, a pose, or a depth map, to a locked text-to-image model. ControlNet copies the encoding blocks of the U-Net and connects the copy to the locked model with 1 × 1 convolutions that start at zero. At the start of training those zero convolutions pass no extra signal, so the locked model is protected. The condition is then able to take over as the new weights learn. The public canny and scribble checkpoints (*lllyasviel/sd-controlnet-canny* and *lllyasviel/sd-controlnet-scribble*) are our two baselines. Canny ControlNet expects a clean three-channel edge map. Scribble ControlNet is the checkpoint we use for freehand drawings. Check-in 1 already recorded the practical gap: on early trials the pretrained model followed clean synthetic edges more reliably than incomplete hand drawings. That gap is the reason to adapt ControlNet rather than only tune guidance.

5.4 LoRA

Hu et al. (2022) adapt a large network by adding a low-rank update to selected weight matrices. A frozen matrix W is used as W + BA, where the inner dimension r is much smaller than the matrix dimensions, so the number of new parameters stays small. The method was introduced for language models. The same pattern is now standard for diffusion models through parameter-efficient fine-tuning. We attach LoRA to the ControlNet attention projections *to_q*, *to_k*, *to_v*, and *to_out.0*, with rank 8, alpha 8, and Gaussian initialization. Only those adapter weights receive gradients. The adapter is what inference reloads with *--lora-path*. Full fine-tuning of ControlNet, or of the U-Net, would touch far more weights than a 50 to 100 sketch set can support, and it would cost more GPU memory on the HPC node. LoRA is the adaptation method we are developing for that reason.

5.5 Synthetic sketches, and the metrics

Paired sketch-photo data at COCO scale does not exist for this project, so the large photo set is paired with edges. Canny (1986) is a classical detector: smooth the image, estimate the gradient, suppress non-maximum responses, and apply two thresholds. Check-in 1 found the expected tradeoff. Lower thresholds

keep thin and broken contours and also keep noise. Higher thresholds make a cleaner map and drop weak detail. Xie and Tu (2015) learn edges with a holistically nested network. Their maps tend to follow object boundaries more continuously than Canny, and they sometimes keep texture and background edges. We will build both Canny and HED maps for the COCO subset. The hand-drawn set remains the test of whether adaptation helps on strokes that neither detector produced.

Heusel et al. (2017) define FID as the Fréchet distance between Inception features of real images and Inception features of generated images. Lower is closer to the real set. FID on only 50 to 100 images is noisy, so we will compute it on the larger COCO-based generation set and treat the small hand-drawn set with CLIP score and the human survey. Radford et al. (2021) train CLIP to embed images and text in one space. CLIP score is the similarity between the generated image and the prompt. It does not measure whether the sketch was followed, which is why the survey and the side-by-side grids stay in the plan. Lin et al. (2014) provide COCO, whose captions are our prompts.

We also considered a conditional GAN such as pix2pix (Isola et al., 2017) as a second family of sketch-to-image models. Pix2pix learns a paired translation from scratch and needs a large paired training set to look photographic. It does not give us a frozen text-to-image prior. We keep it as the alternative we are not pursuing, and we put the project effort into ControlNet plus LoRA.

6. Approach

We have started the overall pipeline in the repository, and we have already compared the two conditioning routes that the baseline needs. The pipeline has four stages.

1. **Data pairing.** Each training example is a target photo, a conditioning image, and a one-line prompt. For the COCO subset the conditioning image is a Canny or HED edge map of the photo, and the prompt is the COCO caption. For the hand-drawn set the conditioning image is the drawing, and the prompt is written with the sketch. Filenames share one stem across *images/*, *sketches/*, and *prompts/*. A checker counts complete triples before a training job starts.
2. **Pretrained baseline.** Stable Diffusion 1.5 and one ControlNet checkpoint (canny or scribble) generate a 512×512 image from the sketch and the prompt. The conditioning scale and the guidance scale are arguments. For canny input, the script can build the edge map with configurable low and high thresholds, or it can accept an edge map that was already built. Each run saves the generated photo, the conditioning image that ControlNet actually saw, and a text record of the prompt, scales, step count, and seed.
3. **LoRA adaptation.** The same frozen Stable Diffusion 1.5 components encode the photo and the prompt. ControlNet receives the noisy latent, the text embedding, and the sketch. LoRA on the ControlNet attention layers is trained with AdamW. The saved adapter is loaded back into the baseline script so the comparison uses one inference path.
4. **Comparison.** The same sketch and prompt are rendered with the pretrained ControlNet and with the adapter. A sweep repeats the pretrained baseline at conditioning scales 0.5, 1.0, and 1.5 and guidance scales 5.0, 7.5, and 12.0. The same grid will be run again after fine-tuning. FID, CLIP score, and the pairwise survey then score the two systems.

The two approaches already exercised in code are the pretrained canny condition and the pretrained scribble condition. Scribble is the route for freehand sketches. Canny is the route for clean edges and for

the threshold study from Check-in 1. LoRA on scribble ControlNet is the third stage, and it is the stage we are developing rather than treating as a finished experiment. Full U-Net fine-tuning and a from-scratch pix2pix model were set aside because both ask a small hand-drawn set to carry a photorealistic generator.

7. Algorithms

The algorithms below are the ones implemented in *src/*. Defaults match the code so the report and the repository describe the same system.

7.1 Shared settings

src/model_setup.py fixes the three identifiers every other script imports: Stable Diffusion 1.5 (*runwayml/stable-diffusion-v1-5*), ControlNet canny (*lllyasviel/sd-controlnet-canny*), ControlNet scribble (*lllyasviel/sd-controlnet-scribble*), and image size 512.

7.2 Baseline inference

src/baseline_controlnet.py loads the selected ControlNet and Stable Diffusion 1.5 in float16 on CUDA and in float32 otherwise. An optional LoRA directory is attached to ControlNet before the pipeline is built. The scheduler is UniPC. The sketch is resized to 512 × 512. If *--apply-canny* is set, Canny edges are computed with default thresholds 100 and 200 and stacked into three channels. The call uses 30 denoising steps, guidance scale 7.5, conditioning scale 1.0, and seed 261 unless the command overrides them. The output directory receives the photo, a *_condition* image, and a text file of the run settings.

7.3 LoRA training

src/train_controlnet_lora.py loads the CLIP tokenizer and text encoder, the variational autoencoder, the U-Net, and the chosen ControlNet from the same checkpoints. The text encoder, autoencoder, and U-Net are frozen and set to evaluation mode. ControlNet receives a LoRA config of rank 8 on *to_q*, *to_k*, *to_v*, and *to_out.0*. The training step encodes the photo to a latent, samples a timestep, adds DDPM noise, and encodes the prompt. ControlNet conditions the U-Net on the sketch. The loss is mean squared error between the U-Net noise prediction and the sampled noise. The optimizer is AdamW with learning rate 1e-4. The default run is 500 steps at batch size 1, conditioning scale 1.0, and seed 261. The script stops immediately if the folder has no complete sketch, photo, and prompt triple, or if LoRA attaches no trainable parameters. The adapter is written under the output directory for that control type.

7.4 Scale sweep and data check

src/run_baseline_ablation.py calls the baseline once for every pair in the grid {0.5, 1.0, 1.5} × {5.0, 7.5, 12.0}. Nine images are the pretrained reference for one sketch before any fine-tuning. *src/check_hand_drawn_data.py* walks *data/hand_drawn/sketches* and reports which stems lack a photo or a *.txt* prompt. Training uses the same pairing rule, so the checker is the preflight for an HPC job.

8. Meaningful check-ins

Our team has nine commits on the main branch. Each commit represents a different part of our Check-in 1 project, including the model setup, ControlNet implementation, LoRA fine-tuning, data organization, testing, and report preparation. These commits show our progress throughout the project. The total of nine commits is within the expected range of approximately 8–10 meaningful check-ins for a three-person team. The table below summarizes each commit and what was added.

Date	Commit	What it added
Sep 27, 2026	bc9328d	Stable Diffusion 1.5 and ControlNet checkpoint names, image size, requirements, and the first README.
Sep 29, 2026	59ed429	Pretrained canny and scribble ControlNet inference for the baseline.
Sep 30, 2026	9f3fe4d	LoRA fine-tuning of ControlNet on hand-drawn sketch, photo, and prompt triples.
Oct 4, 2026	8d0793e	Hand-drawn *images/*, *sketches/*, and *prompts/* folders and the filename rule.
Oct 4, 2026	ccc73f5	A text record of the prompt, scales, steps, and seed saved beside each baseline photo.
Oct 8, 2026	73f10a1	The conditioning-scale and guidance-scale sweep for the pretrained baseline.
Oct 9, 2026	4a77273	A check that every sketch has a matching photo and prompt before training.
Oct 9, 2026	0947c2c	The conditioning image saved beside each baseline photo.
Oct 11, 2026	71f32ec	The Check-in 1 report, *project-check-in-1.pdf*.

Repository link, repeated for the submission requirement: https://github.com/LRozay/Sketch-to-Photo

9. Data and current status

Photos and captions will come from a COCO 2017 subset of about 2,000 to 5,000 images, resized to 512 × 512. Captions are the text prompts. Synthetic sketches for that subset are Canny maps, with the threshold tradeoff already observed in Check-in 1, and HED maps, which gave more continuous object contours and sometimes kept background edges. The hand-drawn collection is 50 to 100 drawings with a short prompt each. The repository has the three folders and the checker. The image files themselves are not committed yet. Leo Rozay owns dataset preparation: the COCO subset, Canny and HED maps, and the hand-drawn collection. Sriyavarma Saripella owns the ControlNet setup and the LoRA training code, which is what *src/* contains now. Venkata Satya Charan Naga Deep Patrini owns FID, CLIP score, the survey, and the report.

Relative to the Check-in 1 next steps, the baseline script, the LoRA script, the ablation grid, and the data check are in the repository. Still ahead of the final report are the finished COCO and hand-drawn files, the HED export, the first HPC training run, and the first recorded FID and CLIP scores. This draft stops at the implemented pipeline and does not fill the results section early.

10. Evaluation plan and remaining work

Once the hand-drawn triples are in *data/hand_drawn*, the HPC job will run the pretrained scribble baseline, train the rank-8 LoRA adapter for the default 500 steps, and reload that adapter on the same sketches. The

nine-cell scale grid will be repeated for the baseline and for the adapter. We will keep the seed fixed so a visual pair differs by the model and the scales, not by a new noise draw. FID will be computed against held-out COCO photographs on the larger generated set. CLIP score will be computed per prompt. The survey will show raters the sketch, the prompt, and the two photos, and ask which photo follows the sketch more closely. Failure cases we already expect, from the Check-in 1 baseline trials and from the literature, are ignored thin strokes, extra objects invented to satisfy the prompt, and copied strokes that hurt realism after fine-tuning on a small set.

The next work, in order, is to finish the synthetic and hand-drawn sets, run the baseline end to end on the HPC, record baseline FID and CLIP score, start the first LoRA run, and fill the results and discussion sections of this outline.

11. References

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